

CLASS-8

LESSON-8 ALGEBRAIC EXPRESSIONS AND INDETTITIES

(This PDF Based on NCERT Book)

ADDITION AND SUBTRACTION OF ALGEBRAIC EXPRESSIONS(बीजीय व्यंजको का जोड़ और घटाव)-

Addition: Collecting Like Items

When you add algebraic expressions, you are simply **combining 'like terms'**.

- Think of terms with the same variable and exponent (like x^2 , y , or just numbers) as being the **same type of object**.
- You can only add things that are the same.

Algebraic Term	Analogy
$4x$ and $3x$	4 Apples and 3 Apples
$5y^2$ and $-2y^2$	5 Blue Blocks and -2 Blue Blocks
6 and 1	6 Dollars and 1 Dollar

The process:

1. **Find the matching items (like terms).**
2. **Add the numbers** in front of those matching items.
3. The item itself (the variable part) **stays the same**.

Example: $(4x + 5) + (3x + 1)$

- **Combine the x items:** $4x + 3x = 7x$ (7 Apples)
- **Combine the number items (constants):** $5 + 1 = 6$ (6 Dollars)
- **Result:** $7x + 6$

Subtraction: The Sign Change Rule

Subtraction works the same way, but you have one **key extra step** before combining.

1. **Change the Signs:** When you subtract an expression, you must **change the sign of every single term** in the expression you are taking away.
2. **Combine Like Terms:** After changing the signs, treat it like an addition problem and combine the like terms.

Example: $(7x + 5) - (3x + 1)$

- Step 1: Change the signs in the second set of parentheses:

$$7x + 5 - 3x - 1$$

- **Step 2: Combine the like items:**
 - **x terms:** $7x - 3x = 4x$
 - **Number terms:** $5 - 1 = 4$
- **Result:** $4x + 4$

MULTIPLICATION OF ALGEBRAIC EXPRESSIONS(बीजीय व्यंजकों का गुणन)-

Multiplication: The Distribute Rule

Think of algebraic multiplication as making sure **every part of the first expression multiplies every part of the second expression**. This is often called the **Distributive Property**.

1. Monomial by Monomial (Single Term by Single Term)

This is the simplest form, where you multiply the parts in three steps:

1. **Multiply the Coefficients (Numbers):** Multiply the large numbers in front of the variables.
2. **Multiply the Variables:**
 - If the variables are **different** (like x and y), just write them next to each other.
 - If the variables are the **same** (like x and x), you **add their exponents**.
3. **Combine the Results.**

Example	Step-by-Step	Result
$(2x) \times (3x^2)$	1. Numbers: $2 \times 3 = 6$	
	2. Variables: $x \times x^2 = x^{(1+2)} = x^3$ (Remember, x is x^1)	$6x^3$
$(-4a^3) \times (5b)$	1. Numbers: $-4 \times 5 = -20$	
	2. Variables: $a^3 \times b = a^3b$ (They are different)	$-20a^3b$

2. Monomial by Polynomial (Single Term by Multiple Terms)

Here, you use the **Distributive Property** to make sure the single term outside the parenthesis multiplies **every single term** inside the parenthesis.

Example: $2x(3x^2 + 4x - 1)$

1. Multiply $2x$ by the first term ($3x^2$):

$$2x \times 3x^2 = 6x^3$$

2. Multiply $2x$ by the second term ($+4x$):

$$2x \times 4x = 8x^2$$

3. Multiply $2x$ by the third term (-1):

$$2x \times (-1) = -2x$$

Result: $6x^3 + 8x^2 - 2x$

3. Polynomial by Polynomial (Multiple Terms by Multiple Terms)

When multiplying two multi-term expressions (like two binomials, which have two terms each), you use the **FOIL** method (First, Outer, Inner, Last) or simply use the **Distributive Property twice**.

Example: $(x + 2)(x - 3)$

1. Take the first term of the first expression (x) and distribute it to the second expression:

$$x(x - 3) = x^2 - 3x$$

2. Take the second term of the first expression ($+2$) and distribute it to the second expression:

$$+2(x - 3) = +2x - 6$$

3. Combine the results and collect like terms:

$$(x^2 - 3x) + (+2x - 6)$$

$$x^2 + (-3x + 2x) - 6$$

$$x^2 - x - 6$$